

needed a way to check more pictures in less time. But simply hiring more image analysis experts would still not be enough to process all seventeen million pictures.

The process of testing is long, complex and manually-intensive. Seagate's Normandale factory takes millions of microscopic pictures every day, generating 10TB of data that must be sifted through to detect potential production defects before wafers are assembled into drives. Because of the sheer volume of transducers that need analysis, engineers cannot possibly test them all. Even with a lengthy manufacturing process, there is simply not enough time to check every image. This means that defective units can and do escape immediate detection occasionally and are caught later in the process, at a much higher cost.

The teams had achieved a degree of automation using rules-based image analysis. This approach identifies some anomalies, but rules had to be built manually, which was a time-consuming process that too is constantly tweaked and refined. The rules-based system was slow to set up, slow to refine and produced variable results. Besides generating plenty of false positives, rules could only detect known issues. This resulted in a potential risk that faulty wafers could escape detection before being assembled into read-and-write heads.

The solution was to deal with two main problems: huge volume of data that needed to be processed every day and shortcomings of the current rules-based analysis engine. Traditional Big Data programs operate on a batch process that is completely inappropriate for a production



line operating 24/7.

However, thanks to advances in artificial intelligence (AI), machine learning and the Internet of Things (IoT) sensors, a new solution took care of this issue. The first step was to build a deep neural network (DNN) that could generate insights to improve automation and detection of transducer failures. Neural network processing was built using Nvidia V100 and P4 GPUs, and high-performance Nytro X 2U24 storage to underpin the deep learning and AI elements.

Wafer images were then fed into the DNN to train the AI system to distinguish between good and bad wafers. It learns in exactly the same way a human engineer does by examining thousands of photographs. But thanks to the raw processing power of the DNN, it learnt much faster and more accurately than a human. Over time, it acquired the ability to spot potential process defects.

Now, it can build and refine its own rules based on anomalies detected during the image analysis operation. Most importantly, it accepts and analyses images generated by the electron microscope in real time. Seagate is now able to process all images

generated every day and can identify tiny defects that may otherwise be missed by a human engineer.

Real-time processing also allows the teams to identify and correct manufacturing issues early. The quicker the problems can be identified, the more effectively Seagate can minimise their impact on the production process and costs.

Project Athena, Seagate's internal AI edge platform, may be excellent at identifying defects but it does not and can not completely replace factory subject matter experts. It opens up new opportunities for wafer experts to innovate and remedy larger problems. Its ability to detect anomalies in a faster, more adaptive and more meaningful way can extend beyond the smart factory and prove useful in domains as varied as public safety, autonomous vehicles and smart cities.

"We expect to deploy Athena across all of our manufacturing facilities in due course," says Jeffrey Nygaard, executive vice president of operations, products and technology, Seagate. And, as the cost of microscopic cameras and the IoT sensors falls, the same technologies can be used for other applications, too.

Seagate's manufacturing tools contain between thirty or more sensors, each of which records machine health and other measurements every second. This information fed into Athena DNN helps identify production issues earlier. In this model, data centre technologies—compute and storage model—are moved to the edge of the network to enable a new generation of applications. **EFY**

—Deepshikha Shukla

Takeaways of the case study

- Seagate has built a working, practical AI platform to improve production line efficiencies and product quality. It could provide up to twenty per cent reduction in new cleanroom investments required for manufacturing and could lead to up to ten per cent reduction in hours spent on the process.
- It can process millions of microscopic photographs every single day.
- By deploying deep learning, Athena has trained itself to identify defects faster and more accurately than a human subject matter expert.
- Seagate can now resolve irregularities and process problems more quickly and at a lower cost than ever before. The company expects to see up to 300 per cent ROI from efficiency improvements and better quality processes.
- This technology has a broad range of applications for the entire manufacturing industry. It is a significant first step in smart manufacturing and an example of Industry 4.0 happening right now.